ICT305 Topics in IT

Topic 6
Data Mining

CIHE Australia

Agenda

1. What is data mining?
2. Data pre-processing
3. Why preprocess the data?
4. Descriptive data summarization
5. Data cleaning
6. Data integration and transformation
7. Data reduction
8. Discretization and concept hierarchy generation

What is data mining?

- After years of data mining there is still no unique answer to this question.

- A tentative definition:

  "Data mining is the use of efficient techniques for the analysis of very large collections of data and the extraction of useful and possibly unexpected patterns in data."

Data Mining

- **Data explosion problem:**
  - Automated data collection tools and mature database technology.
  - Leading to tremendous amounts of data stored in databases, data warehouses and other information repositories.

We are **drowning in data** but starving for knowledge!

Definition of data mining

- **Data mining** (knowledge discovery from data)
  - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
  - Data mining: a misnomer?
- **Alternative names**
  - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- **Watch out:** Is everything "data mining"?
  - (Deductive) query processing.
  - Expert systems or small ML/statistical programs

Definition of data mining

"The process of discovering meaningful new correlations, patterns, and trends by sifting through large amounts of stored data, using pattern recognition technologies and statistical and mathematical techniques."

Real example of data mining

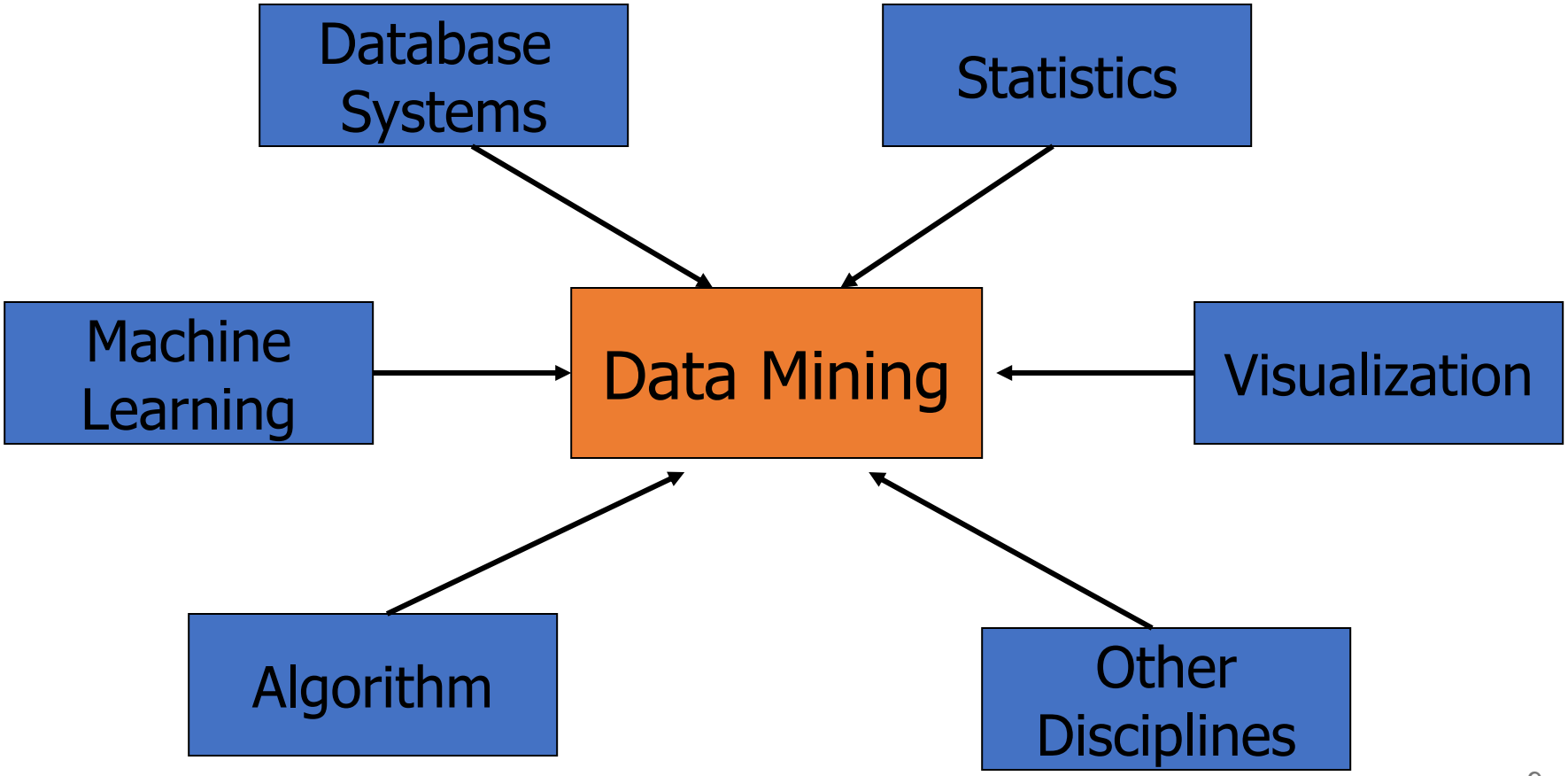
Real Example from the NBA

- Play-by-play information recorded by teams
  - Who is on the court
  - Who shoots
  - Results
- Coaches want to know what works best
  - Plays that work well against a given team
  - Good/bad player matchups
- Advanced Scout (from IBM Research) is a data mining tool to answer these questions.

Why do we need data mining?

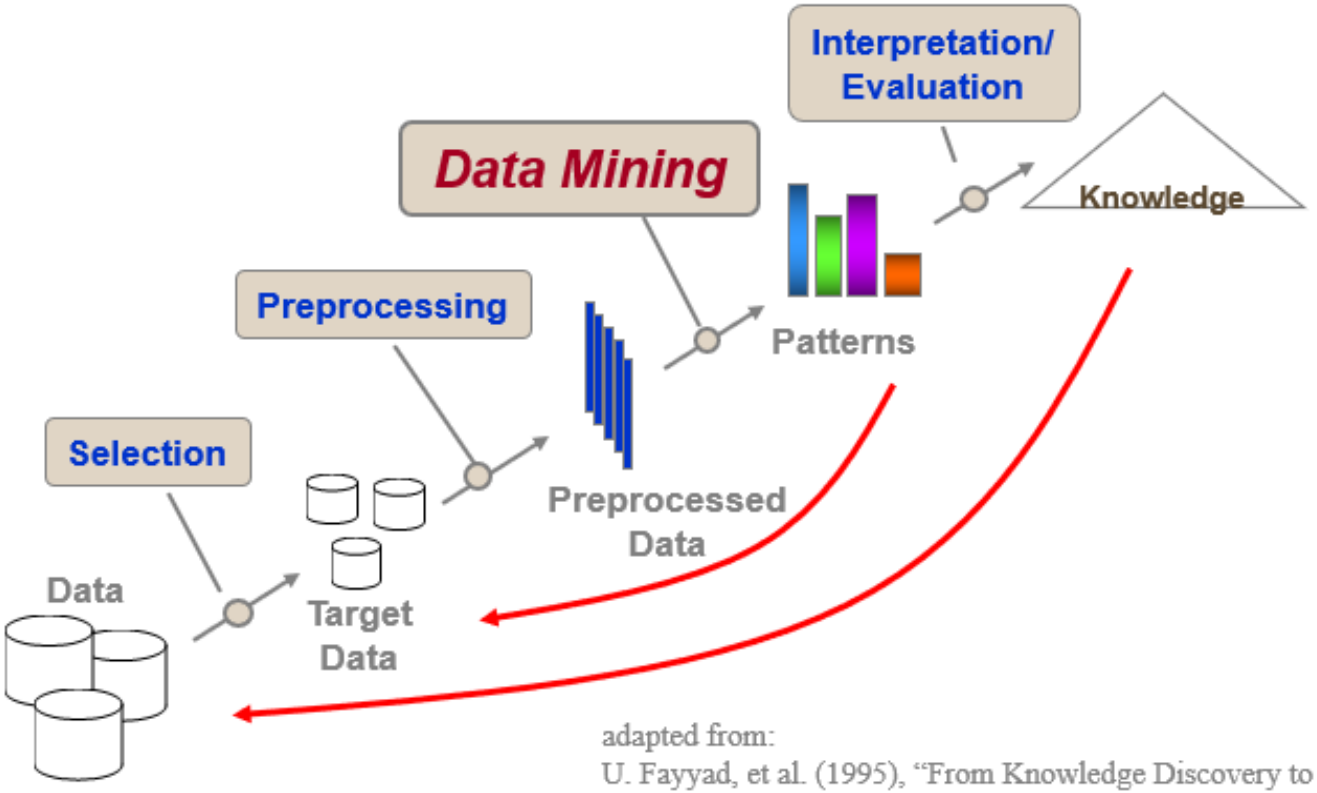
- Really, really huge amounts of raw data!!
  - In the digital age, TB of data is generated by the second
    - Mobile devices, digital photographs, web documents.
    - Facebook updates, Tweets, Blogs, User-generated content
    - Transactions, sensor data, surveillance data
    - Queries, clicks, browsing
  - Cheap storage has made possible to maintain this data
- Need to analyze the raw data to extract knowledge

Disciplines for data mining



Data mining for knowledge discovery from data (KDD)

Process for KDD



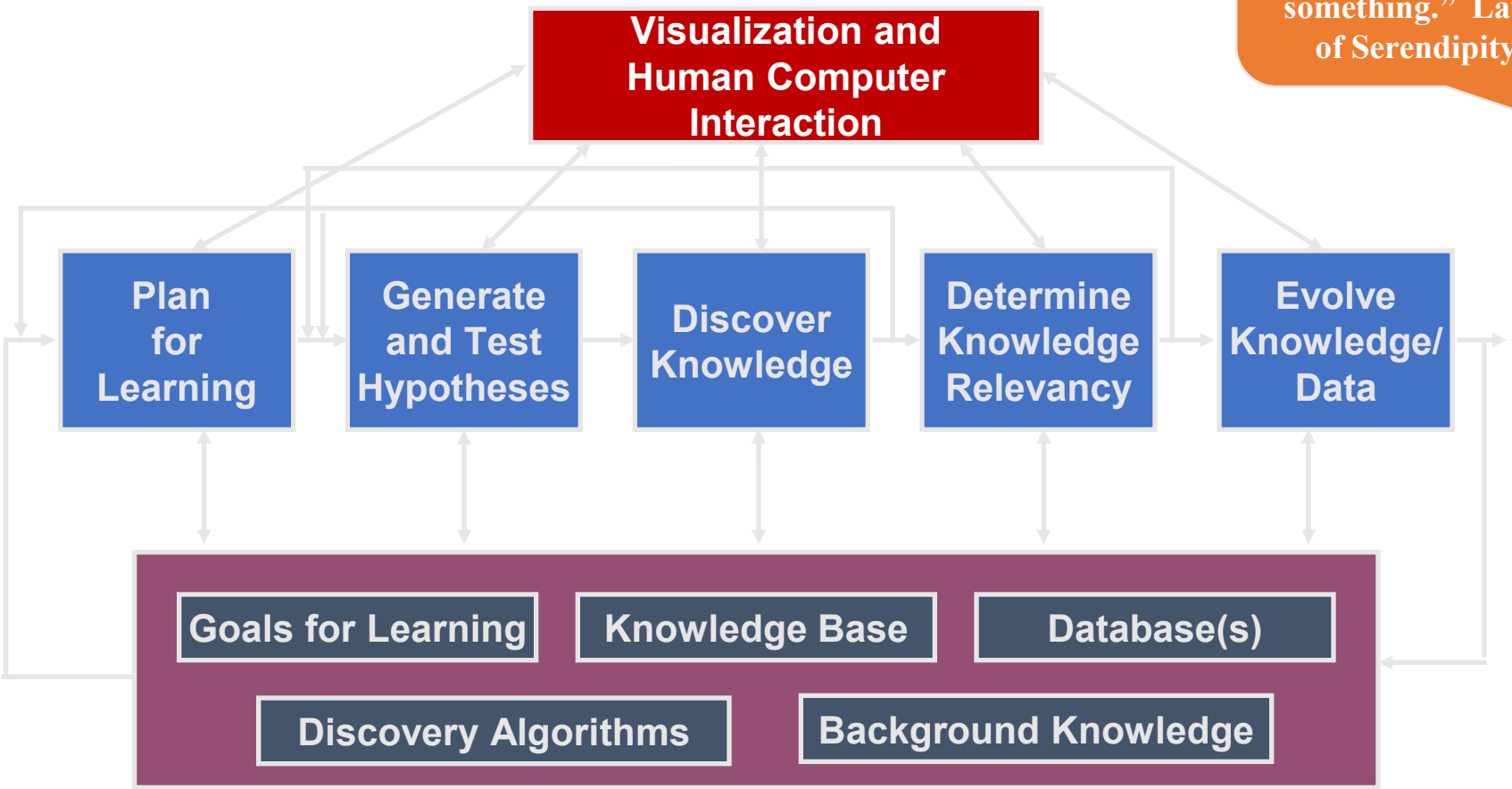
adapted from:
U. Fayyad, et al. (1995), "From Knowledge Discovery to Data Mining: An Overview," Advances in Knowledge Discovery and Data Mining, U. Fayyad et al. (Eds.), AAAI/MIT Press

Multi-dimensional view of data mining

- **Data to be mined**
  - Relational, data warehouse, transactional, stream, object-oriented/relational, active, spatial, time-series, text, multi-media, heterogeneous, legacy, WWW, etc.
- **Knowledge to be mined**
  - Characterization, discrimination, association, classification, clustering, trend/deviation, outlier analysis, etc.
  - Multiple/integrated functions and mining at multiple levels.
- **Techniques utilized**
  - Database-oriented, data warehouse (OLAP), machine learning, statistics, visualization, etc.
- **Applications adapted**
  - Retail, telecommunication, banking, fraud analysis, bio-data mining, stock market analysis, Web mining, etc.

Factors of an effective and successful KDD process for HCI and visualization

"In order to discover anything, you must be looking for something." Laws of Serendipity



Data mining: classification schemes (2 major classes of data mining)

- **General functionality**
  - Descriptive data mining
  - Predictive data mining
- **Different views, different classifications**
  - Kinds of data to be mined
  - Kinds of knowledge to be discovered
  - Kinds of techniques utilized
  - Kinds of applications adapted

Potential applications based on data mining

Market Analysis and Management

- **Where does the data come from?**
  - Credit card transactions, loyalty cards, discount coupons, customer complaint calls, plus (public) lifestyle studies
- **Target marketing**
  - Find clusters of "model" customers who share the same characteristics: interest, income level, spending habits, etc.
  - Determine customer purchasing patterns over time
- **Cross-market analysis**
  - Associations/co-relations between product sales, & prediction based on such association
- **Customer profiling**
  - What types of customers buy what products (clustering or classification)?
- **Customer requirement analysis**
  - identify the best products for different customers
  - predict what factors will attract new customers

Potential applications based on data mining

Corporate Analysis & Risk Management

- **Finance planning and asset evaluation**
  - cash flow analysis and prediction
  - contingent claim analysis to evaluate assets
  - cross-sectional and time series analysis (financial-ratio, trend analysis, etc.)
- **Resource planning**
  - summarize and compare the resources and spending
- **Competition**
  - monitor competitors and market directions
  - group customers into classes and a class-based pricing procedure
  - set pricing strategy in a highly competitive market

Potential applications based on data mining

Fraud Detection & Mining Unusual Patterns

- **Approaches**: Clustering & model construction for frauds, outlier analysis
- **Applications**: Health care, retail, credit card service, telecoms.
  - ***Auto insurance***: ring of collisions
  - ***Money laundering***: suspicious monetary transactions
  - ***Medical insurance***
    - Professional patients, ring of doctors, and ring of references
    - Unnecessary or correlated screening tests
  - ***Telecommunications***: phone-call fraud
    - Phone call model: destination of the call, duration, time of day or week. Analyze patterns that deviate from an expected norm
  - ***Retail industry***
    - Analysts estimate that 38% of retail shrink is due to dishonest employees

So, what is data?

- Collection of data objects and their attributes.
- An attribute is a property or characteristic of an object
  - *Examples*: eye color of a person, temperature, etc.
  - **Attribute** is also known as *variable, field, characteristic*, or *feature*.
- A collection of attributes describe an object.
  - Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Size: Number of objects
Dimensionality: Number of attributes
Sparsity: Number of populated object-attribute pairs

Types of Attributes

- There are different types of attributes
  - Categorical
    - Examples: eye color, zip codes, words, rankings (e.g, good, fair, bad), height in {tall, medium, short}
  - Numeric
    - Examples: dates, temperature, time, length, value, count.
    - Discrete (counts) vs Continuous (temperature)
    - *Special case*: Binary attributes (yes/no, exists/not exists)

Numeric Record Data

- If **data objects** have the same **fixed set** of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute.
- Such data set can be represented by an n-by-d data matrix, where there are n rows, one for each object, and d columns, one for each attribute.

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Categorical Data

- Data that consists of a collection of records, each of which consists of a fixed set of categorical attributes.

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	High	No
2	No	Married	Medium	No
3	No	Single	Low	No
4	Yes	Married	High	No
5	No	Divorced	Medium	Yes
6	No	Married	Low	No
7	Yes	Divorced	High	No
8	No	Single	Medium	Yes
9	No	Married	Medium	No
10	No	Single	Medium	Yes

Document Data

- Each document becomes a `*term*' vector,
  - *each term* is a *component* (attribute) of the vector,
  - the value of each component is the number of times the corresponding term occurs in the document.
  - Bag-of-words representation – no ordering

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Transaction Data

- Each record (transaction) is a set of items.

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

- A set of items can also be represented as a binary vector, where each attribute is an item.
- A document can also be represented as a set of words (no counts)

Sparsity: percentage of zero values in the transaction-item binary matrix

Ordered Data

- Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

- Data is a long-ordered string

Data Mining Tasks...

- Classification [Predictive]
- Clustering [Descriptive]
- Association Rule Discovery [Descriptive]
- Sequential Pattern Discovery [Descriptive]
- Regression [Predictive]
- Deviation Detection [Predictive]

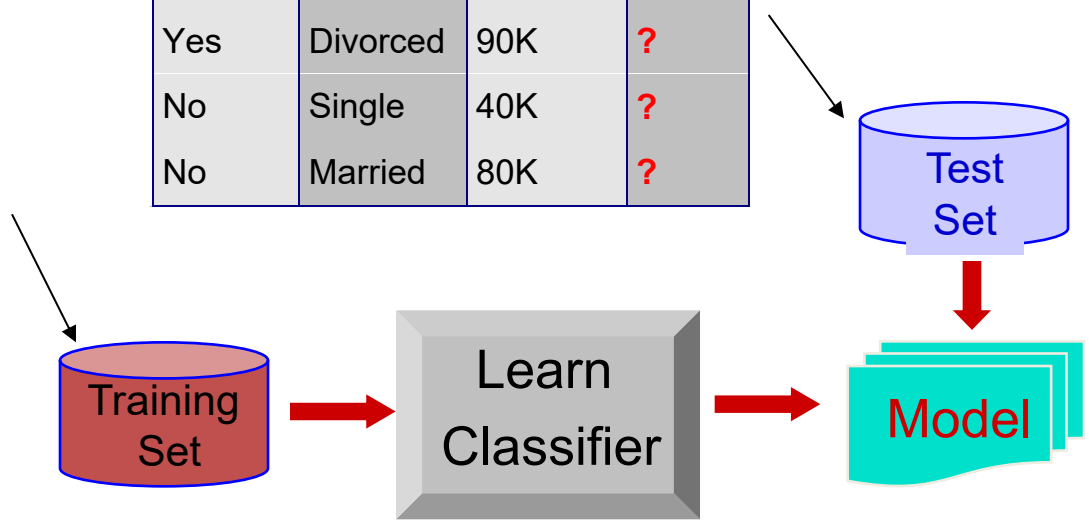
Classification: Definition

- Given a collection of records (*training set*)
  - *Each record* contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- **Goal**: previously unseen records should be assigned a class as accurately as possible.
  - A *test set* is used to determine the accuracy of the model.
  - Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

Classification Example

Tid	Refund	Marital Status	Taxable Income	Cheat
	categorical	categorical	continuous	class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Refund	Marital Status	Taxable Income	Cheat
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?



Classification: Application

- **Fraud Detection**
  - ***Goal***: Predict fraudulent cases in credit card transactions.
  - ***Approach***:
    - Use credit card transactions and the information on its account-holder as attributes.
      - When does a customer buy, what does he buy, how often he pays on time, etc.
    - Label past transactions as fraud or fair transactions. This forms the class attribute.
    - Learn a model for the class of the transactions.
    - Use this model to detect fraud by observing credit card transactions on an account.

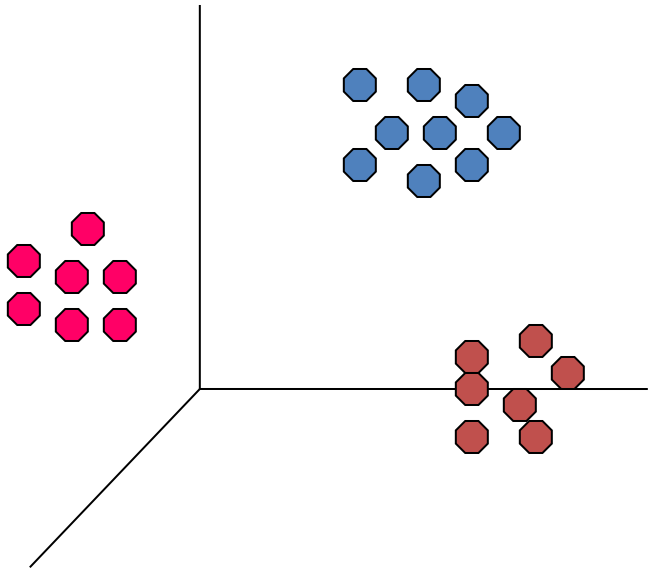
Clustering Definition

- Given a set of data points, each having a set of attributes, and a similarity measure among them, find clusters such that:
  - Data points in one cluster are more similar to one another.
  - Data points in separate clusters are less similar to one another.
- **Similarity Measures**:
  - **Euclidean Distance** if attributes are continuous.
  - Other Problem-specific Measures.

Clustering Definition

Intra-cluster distances are minimized

Intercluster distances are maximized



Clustering: Application

- **Market Segmentation:**
  - ***Goal***: subdivide a market into distinct subsets of customers where any subset may conceivably be selected as a market target to be reached with a distinct marketing mix.
  - ***Approach:***
    - Collect different attributes of customers based on their geographical and lifestyle related information.
    - Find clusters of similar customers.
    - Measure the clustering quality by observing buying patterns of customers in same cluster vs. those from different clusters.

Challenges of Data Mining

- Scalability
- Dimensionality
- Complex and Heterogeneous Data
- Data Quality
- Data Ownership and Distribution
- Privacy Preservation
- Streaming Data

The data analysis pipeline

- Mining is not the only step in the analysis process

- Preprocessing: real data is noisy, incomplete and inconsistent. Data cleaning is required to make sense of the data
  - Techniques: Sampling, Dimensionality Reduction, Feature selection.
  - A dirty work, but it is often the most important step for the analysis.
- Post-Processing: Make the data actionable and useful to the user
  - Statistical analysis of importance
  - Visualization.
- Pre- and Post-processing are often data mining tasks as well.

Data Quality

- Examples of data quality problems:
  - Noise and outliers
  - Missing values
  - Duplicate data

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
9	No	Single	90K	No

A mistake or a millionaire?

Missing values

Inconsistent duplicate entries

Why Data Preprocessing?

- Data in the real world is <u>dirty</u>
  - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - e.g., occupation=“ ”
  - noisy: containing errors or outliers
    - e.g., Salary=“-10”
  - inconsistent: containing discrepancies in codes or names, e.g.,
    - Age=“42” Birthday=“03/07/1997”
    - Was rating “1,2,3”, now rating “A, B, C”
    - discrepancy between duplicate records

Why Is Data Dirty?

- Incomplete data may come from
  - "Not applicable" data value when collected
  - Different considerations between the time when the data was collected and when it is analyzed.
  - Human/hardware/software problems
- Noisy data (incorrect values) may come from
  - Faulty data collection instruments
  - Human or computer error at data entry
  - Errors in data transmission
- Inconsistent data may come from
  - Different data sources
  - Functional dependency violation (e.g., modify some linked data)
- Duplicate records also need data cleaning

Why is Data Preprocessing Important?

- No quality data, no quality mining results!
  - Quality decisions must be based on quality data
    - e.g., duplicate or missing data may cause incorrect or even misleading statistics.
  - Data warehouse needs consistent integration of quality data.
- Data extraction, cleaning, and transformation comprises the majority of the work of building a data warehouse.

Multi-Dimensional Measure of Data Quality

- A well-accepted multidimensional view:
  - Accuracy
  - Completeness
  - Consistency
  - Timeliness
  - Believability
  - Value added
  - Interpretability
  - Accessibility
- Broad categories:
  - Intrinsic, contextual, representational, and accessibility

Major Tasks in Data Preprocessing

- **Data cleaning**
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration**
  - Integration of multiple databases, data cubes, or files
- **Data transformation**
  - Normalization and aggregation
- **Data reduction**
  - Obtains reduced representation in volume but produces the same or similar analytical results
- **Data discretization**
  - Part of data reduction but with particular importance, especially for numerical data

Outliers

- Outliers
  - Extreme values that are separated from the rest of the data.
  - Outliers affect the mean.
- The 1.5 × IQR(Interquartile Range) Rule for Outliers
  - The Interquartile Range is calculated as the difference between the 75th percentile (Q3) and the 25th percentile (Q1) of the data.
  - Call an observation an outlier if it falls more than 1.5 × IQR above the third quartile or below the first quartile.

$$X < Q1 - 1.5 \times IQR$$

$$X > Q3 + 1.5 \times IQR$$

Outliers

In the New York travel time data, we found $Q_1 = 15$ minutes, $Q_3 = 42.5$ minutes, and $IQR = 27.5$ minutes.

- For these data,

$$1.5 \times IQR = 1.5(27.5) = 41.25$$

$$Q_1 - 1.5 \times IQR = 15 - 41.25 = \mathbf{-26.25} \ (\textit{near}\ 0)$$

$$Q_3 + 1.5 \times IQR = 42.5 + 41.25 = \mathbf{83.75} \ (\sim 80)$$

- Any travel time close to 0 minutes or longer than about 80 minutes is considered an outlier.

Data Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values.
- Methods
  - *Smoothing*: Remove noise from data
  - *Attribute/feature construction*
    - New attributes constructed from the given ones
  - *Aggregation*: Summarization, data cube construction
  - *Normalization*: Scaled to fall within a smaller, specified range
    - min-max normalization
    - z-score normalization
    - normalization by decimal scaling
  - *Discretization*: Concept hierarchy climbing

Normalization

- **Min-max normalization**: to [new_min$_A$, new_max$_A$]

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

  - Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,000 is mapped to 0.7093

- **Z-score normalization** (μ: mean, σ: standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

Ex. Let μ = 54,000, σ = 16,000. Then

(73,000 - 54,000) / 16,000 = 1.1875

- **Normalization by decimal scaling**

$$v' = \frac{v}{10^j}$$

Where *j* is the smallest integer such that Max(|v'|) < 1

j = 5 (since 98,000 / 10^5 = 0.98 < 1)

\$73,000 normalized: **73,000 / 100,000 = 0.73**

Conclusions